

# HELPSs (Helicopter Emergency Landing Place Suggestion System)

## Conceptual Architecture and Landing Region Design

**Version:** 1.0

**Purpose:** Establish the conceptual foundation and architectural philosophy of HELPSs before implementation.

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## 1. Introduction

HELPSs (Helicopter Emergency Landing Place Suggestion System) is a **Geospatial Decision Support System** that automatically analyzes a geographic region and identifies potential emergency landing locations for helicopters.

The primary objective of HELPSs is **not to make landing decisions**, but to **assist the pilot** by identifying and ranking the safest possible landing options based on terrain characteristics, helicopter constraints, and environmental conditions.

The pilot always remains the final decision maker.

Unlike navigation systems that focus on reaching destinations, HELPSs focuses on answering a completely different question:

**"If an emergency occurs at this location, where are the safest places a helicopter could land?"**

To answer this question, HELPSs combines satellite imagery, Digital Elevation Models (DEM), GIS data, and helicopter operational constraints to build a geospatial intelligence database.

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## 2. Decision Support Philosophy

A fundamental design philosophy of HELPSs is that it is **not an autonomous landing system**.

The system should never command a pilot to land at a specific location.

Instead, HELPSs performs extensive geospatial analysis and provides:

- Candidate landing locations

- Safety measurements
- Risk factors
- Suitability score
- Reasons for recommendation

For example,

Instead of saying

"Land here."

HELPSs should say

"Among all analyzed locations, this region has the highest suitability score because it satisfies terrain, obstacle, and accessibility requirements."

This philosophy makes the system transparent, explainable, and trustworthy.

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### 3. The Core Idea Behind HELPSs

Initially, it is tempting to think that the project is about finding landing locations.

However, after carefully analyzing the problem, the actual objective becomes much clearer.

HELPSs is fundamentally trying to build a **Geospatial Intelligence Database**.

The database should contain every potential landing region within the Area of Interest (AOI), irrespective of a specific helicopter.

Only after this database has been built should helicopter-specific evaluation begin.

Therefore, HELPSs consists of two major stages.

#### Stage 1

Build a helicopter-independent database.

#### Stage 2

Evaluate the database according to helicopter specifications.

This separation greatly improves scalability because the same database can be reused for different helicopters without repeating feature extraction.

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## 4. What Exactly is a Landing Region?

A Landing Region is the **fundamental entity** of HELPSs.

It is defined as:

**A continuous open geographical area that satisfies the minimum physical requirements to be considered a potential emergency landing location.**

Notice several important aspects of this definition.

A Landing Region is **not**

- a helicopter
- a landing point
- a score
- a grid cell

Instead, it is simply an area that deserves further investigation.

For example,

A large grassland

A dry riverbed

An open field

A barren plateau

can all become Landing Regions.

The purpose of a Landing Region is to represent a continuous piece of land that can later be evaluated for different helicopter types.

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## 5. Is a Landing Region Always a Polygon?

Yes.

The most appropriate representation of a Landing Region is a **polygon**.

Nature does not divide land into squares.

Open areas have irregular boundaries.

A polygon accurately captures:

- Actual boundary
- Shape
- Orientation
- Area
- Perimeter

This is also the standard representation used in Geographic Information Systems (GIS).

Examples include:

- Lakes
- Forests
- Buildings
- Agricultural fields
- Administrative boundaries

Therefore, representing Landing Regions as polygons makes HELPSs compatible with modern GIS tools such as QGIS and PostGIS.

Each Landing Region will contain:

- Region ID
- Polygon Geometry
- Bounding Box
- Centroid
- Area
- Perimeter

This polygon becomes the permanent identity of that land.

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## 6. How Do We Generate Landing Regions?

This is one of the most important questions in the project.

Initially, it may seem reasonable to divide the Area of Interest into grid cells.

Although simple, grid-based approaches introduce artificial boundaries.

Nature is continuous.

Therefore, HELPSs should identify **natural open regions** rather than artificial grid cells.

The proposed pipeline is:

Satellite Imagery



Land Cover Classification



Open Land Detection



Connected Component Analysis



Polygon Generation



Landing Region Database

The key idea is Connected Component Analysis.

Suppose the classified image identifies grass pixels.

Instead of treating every pixel separately, neighbouring connected grass pixels are grouped into one continuous region.

That connected region becomes a Landing Region.

Thus, the geometry of nature itself defines the region rather than an arbitrary grid.

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## 7. Permanent Identity of a Landing Region

Every Landing Region requires a permanent identity.

The identity is created using:

- Region ID
- Polygon Geometry

- Geographic Coordinates
- Creation Timestamp

For example,

Region ID:

LR\_000127

Polygon:

(latitude, longitude) coordinate list

Centroid:

(13.7438, 75.6289)

Area:

842 m<sup>2</sup>

Whenever new satellite imagery becomes available, HELPSs compares newly detected polygons against previously stored polygons.

If they overlap significantly, the existing region is updated.

If an entirely new clearing appears, a new Region ID is generated.

This enables the database to evolve while maintaining historical consistency.

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## 8. Static and Dynamic Information

Not all information changes at the same rate.

Therefore, HELPSs separates information into two categories.

### Static Information

Changes rarely.

Examples:

- Terrain
- Slope

- Elevation
- Roads
- Buildings
- Land Geometry
- Surface Type

Static information is stored permanently.

It may only require updates every few months.

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## Dynamic Information

Changes frequently.

Examples:

- Weather
- Rainfall
- Visibility
- Wind Speed
- Wind Direction
- Temperature

Dynamic information should **not** be permanently stored inside the Landing Region database.

Instead, it is retrieved during evaluation.

This separation keeps the database lightweight while ensuring recommendations remain up to date.

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## 9. What Features Belong to a Landing Region?

Features can be divided into three categories.

### Geometry Features

These describe the physical shape.

- Area
- Perimeter
- Maximum Clear Length
- Maximum Clear Width
- Shape Quality

## Terrain Features

These describe the ground.

- Terrain Slope
  - Terrain Roughness
  - Elevation
  - Surface Type
  - Surface Firmness
  - Water Presence
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## Environmental Features

These describe surrounding conditions.

- Tree Density
- Building Density
- Obstacle Density
- Tower Presence
- Wire Risk
- Distance to Road
- Accessibility

These measurements remain independent of helicopter type.

They simply describe the region.

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# 10. Which Features Vary Within the Same Region?

One important realization during project design is that a Landing Region is rarely uniform.

Consider a large open field.

One side may be perfectly flat.

Another side may contain a steep slope.

Similarly,

Tree density,

Roughness,

Obstacle proximity,

can all vary across the same polygon.

Therefore,

features such as

- slope
- roughness
- obstacle density

should not always be represented by a single average value.

Instead,

HELPSs should preserve their spatial variation whenever possible.

For Version 1, summary statistics such as maximum, minimum, mean, and standard deviation may be stored.

Future versions can maintain higher-resolution spatial measurements for more accurate evaluation.

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## 11. What Should Be Stored Permanently?

The database should only store information that is expensive to compute and changes infrequently.

Examples include:

Region Information

- Region ID
- Polygon
- Area
- Perimeter
- Bounding Box
- Centroid

Terrain Information

- Elevation
- Average Slope
- Maximum Slope
- Terrain Roughness

#### Surface Information

- Surface Type
- Vegetation Density
- Water Presence

#### Infrastructure

- Road Distance
- Building Density
- Tower Locations

This database represents the permanent geospatial intelligence of the Area of Interest.

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## 12. What Should Be Computed During Query Time?

Certain values depend on the selected helicopter and current environmental conditions.

These should never be permanently stored.

Examples include:

- Final Suitability Score
- Weather Score
- Wind Penalty
- Mission Priority
- Helicopter Compatibility
- Overall Ranking

This makes HELPSs flexible enough to support future helicopter types without rebuilding the database.

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## 13. How Does the Evaluation Engine Work?

After the database has been constructed, helicopter evaluation begins.

Every helicopter has unique operational characteristics.

Examples include:

- Rotor Diameter
- Overall Length
- Landing Footprint
- Maximum Allowable Slope

- Rotor Clearance
- Safety Buffer

When a helicopter is selected, HELPSs retrieves all Landing Regions from the database.

Each region is evaluated against helicopter requirements.

For example,

If a Landing Region has

Maximum Clear Diameter = 18 m

then

RPAS

may satisfy this requirement,

whereas

ALH

may not.

Similarly,

a region with

Slope = 5°

may be acceptable for one helicopter but exceed another helicopter's operating limitation.

Therefore,

the same Landing Region can receive different suitability scores depending on the selected helicopter.

This is the key reason why the database must remain helicopter independent.

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## 14. Final Suitability Score

The final recommendation should combine multiple sources of information.

Conceptually,

Final Suitability = Static Suitability + Dynamic Environment + Mission Context

where

Static Suitability represents terrain characteristics,

Dynamic Environment represents current weather,

Mission Context represents operational priorities.

This modular design allows the same database to support various emergency scenarios without modification.

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## 15. Conclusion

The most important realization during the conceptual design of HELPSs is that the project is **not about helicopters first**.

It is about **understanding the geography first**.

HELPSs therefore builds a helicopter-independent geospatial intelligence database containing every potential Landing Region within the Area of Interest.

Only after this intelligence has been created does the system evaluate regions according to helicopter-specific operational constraints and real-time environmental conditions.

This separation between **geospatial intelligence** and **aircraft evaluation** provides scalability, explainability, maintainability, and future extensibility.

It transforms HELPSs from a simple landing recommendation prototype into a reusable geospatial decision support framework capable of supporting multiple helicopter platforms and future operational enhancements.